

Assignment No- 01

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Answer to the Question No.-1:

Central dogma is a important concept in molecular biology. It explains the flow of genetic information within a cell. This includes three main polymers. They are deoxyribonucleic acid (DNA), ribonucleic acid (RNA) and proteins. A shoer overview of thei specific monomers and bonds type is given below:

- i. DNA (Deoxyribonucleic Acid): The monomers of DNA are nucleotides. Each of the nucleotides contains three components, a phosphate group, a deoxyribose sugar and one of four nitrogen bases. Which are adenine, guanine, cytosine and thymine. Nucleotides in DNA are connected through phosphodiester bonds. These bonds form between the carbon of one nucleotide's sugar molecule and the carbon of the next nucleotide's sugar molecule. The specific pairing of the nitrogenous bases (adenine with thymine, and cytosine with guanine). This double-stranded helical structure is known as the DNA double helix.
- ii. RNA (Ribonucleic Acid): The monomers of RNA are nucleotides as well. Each nucleotide consists of a phosphate group, a ribose sugar molecule, and one of four nitrogenous bases (adenine, uracil, cytosine, or guanine). Similar to DNA, the nucleotides in RNA are joined by phosphodiester bonds, but unlike DNA, RNA is single stranded.
- iii. Proteins: Amino acids are the monomers of proteins. There are 20 different types of amino acids. Aminos acids consist an amino group, a carboxyl group and a side chain (R-group). Amino acids are linked together through peptide bonds. The sequence of amino acids determines the primary structure of a protein.

In DNA and RNA, the phosphodiester bonds create a linear backbone. While the hydrogen bonds between complementary nitrogenous bases allow for the formation of double-stranded DNA or RNA secondary structures. These structures are essential for DNA replication, transcription, and translation processes. This type of bonds involved in each polymer influences their structure and function. Moreover, in proteins, the peptide bonds between amino acids form a linear polypeptide chain. The sequence of amino acids and the interactions between the side chains determine the protein's three-dimensional structure, folding, and functional properties. So it can be said that the fundamental principles of molecular biology are based on the unique monomers and bonds found in DNA, RNA, and proteins, which are essential for the transmission and expression of genetic information.

Answer to Question No.-02:

- i. In the question a marine food web is given. One food chain from the above food web is-

Plankton → Krill → Mackerel → Tuna

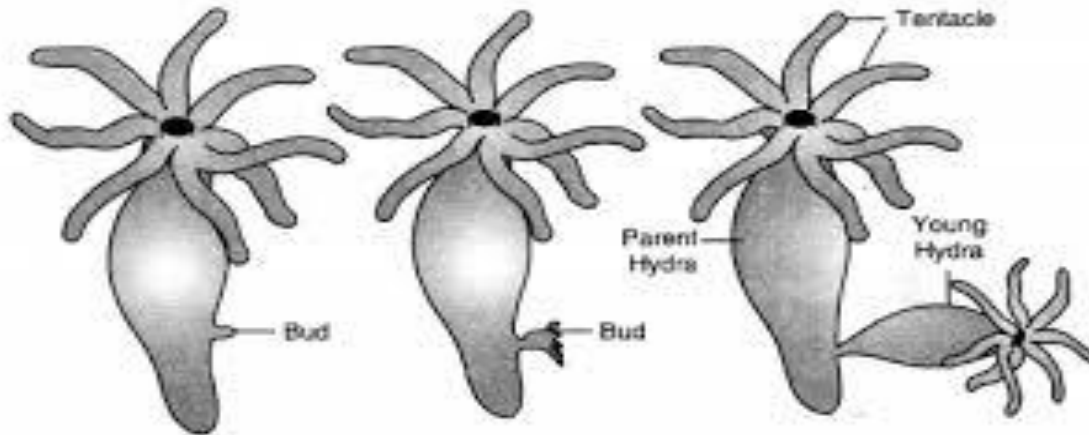
In this four organism food chain Plankton is the producer and rest of them are consumers of their respective level as they cannot produce food on their own. Plankton is a microscopic organism that lives in the ocean. It captures sunlight and uses photosynthesis to create energy. This energy is the base of the food chain. As the consumers cannot produce energy from the sun, so they fully depend on producers like plankton directly or indirectly.

- ii.

A food chain is important to an ecosystem because it shows the transfer of energy from one organism to another. Sunlight energy is collected by producers, who subsequently pass it on to consumers through their consumption of the producers. When consumers eat other consumers, more energy is exchanged. Energy moves from the sun to the top predators in the ecosystem in this way. So, every organism is connected with each other. For example, if the population of plankton decreases, this may cause the population of Krill to fall as well since Krill consume plankton. The number of cod, that eat herring, may then drop as a result, and so on. It shows how important every organism of the food chain is connected with each other.

Answer to Question No.-03

Hydra uses asexual reproduction process to create its offspring which is known as budding. As it's an asexual process, so mitosis occurs during this budding process. Mitosis is a form of cell division that results in the formation of two identical daughter cells, each containing the same number of chromosomes as the parent cell. Mitosis is responsible for the growth, development and asexual reproduction of organisms. In this budding process, the bud is mainly an outgrowth from the body of the parent Hydra and it develops into a genetically identical offspring. This growth is happening because of mitosis cell division. In mitosis, the genetic material of the parent cell is duplicated, and the replicated chromosomes are distributed equally into two daughter cells. It ensures that each daughter cell receives a complete set of chromosomes. This process allows for the production of genetically identical offspring, because the genetic material is faithfully replicated and distributed. On the other hand, meiosis involves the formation of haploid cells also known as gametes. The number of chromosomes is also halved in the daughter cell from each parent cell. In the case of Hydra budding, the goal is to produce a genetically identical offspring, making mitosis the appropriate type of cell division for this process.



Reference

Geo. W. Tannreuther. (1909). Budding in Hydra. *Biological Bulletin*, 16(4), 210–214.

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